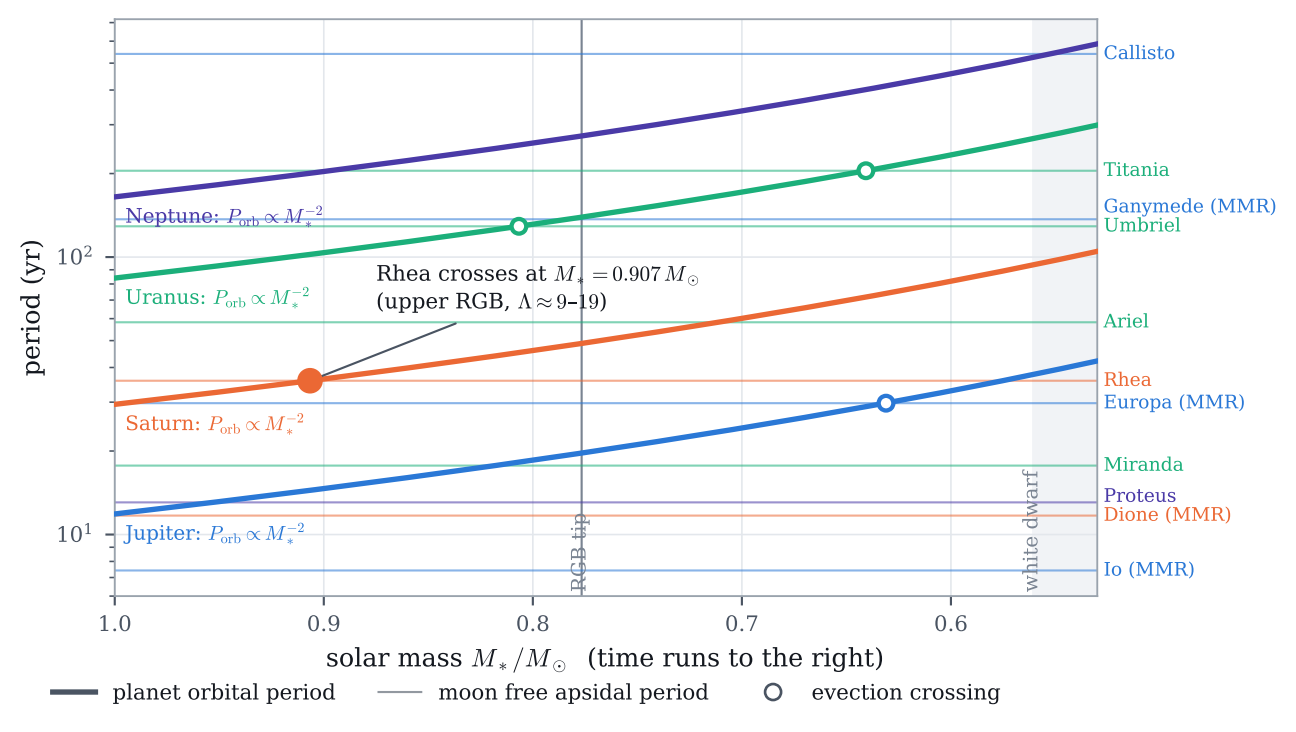
period (yr)
10^2
10^1
Neptune: $P_{\rm orb} \propto M_*^{-2}$
Uranus: $P_{\rm orb} \propto M_*^{-2}$
Saturn: $P_{\rm orb} \propto M_*^{-2}$
Jupiter: $P_{\rm orb} \propto M_*^{-2}$
Rhea crosses at $M_* = 0.907\,M_\odot$
(upper RGB, $\Lambda \approx 9$–19)
RGB tip
white dwarf
Callisto
Titania
Ganymede (MMR)
Umbriel
Ariel
Rhea
Europa (MMR)
Miranda
Proteus
Dione (MMR)
Io (MMR)
1.0
0.9
0.8
0.7
0.6
solar mass $M_*/M_\odot$ (time runs to the right)
planet orbital period
moon free apsidal period
evection crossing